Assignment 3

CS 6117 (Fall 2025)

Exercise 1. Let $\mathbf{C}$ be a locally small category, and let X and Y be objects of $\mathbf{C}$. Consider the functor $F(-) = \mathbf{C}(-, X) \times \mathbf{C}(-, Y) : \mathbf{C}^{op} \to \mathbf{Set}$.

1. Write down what it means for F to be represented by an object Z.
2. By the Yoneda lemma, the natural isomorphism α from Part 1 determines a universal element for F, i.e., the element $x_\alpha \in F(Z)$ corresponding to the natural transformation α under the Yoneda lemma. Describe this element x_α in terms of α.
3. We have seen the object Z before many times in this class: what is it? Prove that it satisfies the required universal property.

Other universal properties can be defined in a similar way.

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1. For F to be represented by an object Z means there exists $Z \in \mathbf{C}$ such that there is a natural isomorphism $\alpha : \mathbf{C}(-,) \xrightarrow{\sim} F(-)$. That is for every $A \in \mathbf{C}$, $\alpha_A : \mathbf{C}(A, Z) \to \mathbf{C}(A, X) \times \mathbf{C}(A, Y)$ is a bijection natural in A. For any morphism $f : B \to A$ in $\mathbf{C}$, the diagram commutes:

$$\begin{array}{ccc} \mathbf{C}(A, Z) & \xrightarrow{\alpha_A} & \mathbf{C}(A, X) \times \mathbf{C}(A, Y) \\ {\scriptstyle \mathbf{C}(f,Z)}\downarrow & & \downarrow{\scriptstyle F(f)} \\ \mathbf{C}(B, Z) & \xrightarrow[\alpha_B]{} & \mathbf{C}(B, X) \times \mathbf{C}(B, Y) \end{array}$$

   where $F(f) = \mathbf{C}(f, X) \times \mathbf{C}(f, Y)$ acts by post-composition: $(g, h) \mapsto (g \circ f, h \circ f)$.

2. By Yoneda lemma, $Nat(\mathbf{C}(-, Z), F) \cong F(Z)$, where a natural transformation α corresponds to the element $x_\alpha = \alpha_Z(\mathrm{id}_Z) \in F(Z)$. Thus, in terms of α, the universal element is $x_\alpha = \alpha_Z(\mathrm{id}_Z) \in \mathbf{C}(Z, X) \times \mathbf{C}(Z, Y)$.

3. The object Z is the product $X \times Y$ in $\mathbf{C}$. The product has projections $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$, such that for any object A and morphisms $f : A \to X$, $g : A \to Y$, there exists a unique morphism $\langle f, g \rangle : A \to X \times Y$ satisfying $\pi_X \circ \langle f, g \rangle = f$ and $\pi_Y \circ \langle f, g \rangle = g$. Define

this is fine but a bit different: we asked to show that if Z is representing, then it is the product. but you've proved the other direction: if Z is the product, then it is representing.

the transformation $\alpha_A : \mathbf{C}(A, X \times Y) \to \mathbf{C}(A, X) \times \mathbf{C}(A, Y)$ by $\alpha_A(h) = (\pi_X \circ h, \pi_Y \circ h)$. This is bijective by the universal property of products and natural because for $k : B \to A$, $\alpha_B(h \circ k) = (\pi_X \circ h \circ k, \pi_Y \circ h \circ k) = F(k)(\alpha_A(h))$. Thus, $\mathbf{C}(-, X \times Y) \cong F(-)$ naturally, so F is represented by $Z = X \times Y$.

Exercise 2. Let **Cat** be the category of all small categories (i.e., categories where the collection of objects forms a set) and functors between small categories.

1. In any small category, the collection of all morphisms also forms a set. Define a morphism functor $M : \mathbf{Cat} \to \mathbf{Set}$ that sends a category to its set of morphisms.

2. Prove that M is representable. What is a representing object for this functor? [Hint: one possibility is a (very) small category.]

Functors from a small category can be used to detect shapes in other categories. For instance, the functor $T : \mathbf{Cat} \to \mathbf{Set}$ that sends a category $\mathbf{C}$ to the set of all commuting triangles in $\mathbf{C}$ is also representable. Taken together, these functors can provide a view of a category as some kind of abstract geometric space, where objects are points (0-cells), morphisms are edges (1-cells), and commuting triangles are faces (2-cells); see nerve of a category.

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1. Since the objects form a set and $\mathbf{C}$ is locally small, the colection of all morphisms $Mor(\mathbf{C}) = \coprod_{A,B \in \mathrm{Ob}(\mathbf{C})} \mathbf{C}(A, B)$. Then, on objects, for a small category $\mathbf{C}$, $M(\mathbf{C}) = Mor(\mathbf{C})$, the set of all morphisms in $\mathbf{C}$. On morphisms, for a functor $F : \mathbf{C} \to \mathbf{D}$, $M(F) : Mor(\mathbf{C}) \to Mor(\mathbf{D})$ is the function sending each morphism f in $\mathbf{C}$ to $F(f)$ in $\mathbf{D}$. This is as a morphism functor $M : \mathbf{Cat} \to \mathbf{Set}$

2. M is representable. Think the representing object is the small category $\mathbf{2}$ with objects $\{0, 1\}$, morphisms $\{\mathrm{id}_0, \mathrm{id}_1, a : 0 \to 1\}$. To prove that, consider there is a natural isomorphism $\alpha : \mathbf{Cat}(\mathbf{2}, -) \xrightarrow{\sim} M(-)$. For a small category $\mathbf{C}$, a functor $F : \mathbf{2} \to \mathbf{C}$ correspond to choosing $A = F(0)$, $B = F(1)$,for some $A, B \in \mathbf{C}$. And $F(\mathrm{id}_0) = \mathrm{id}_A$, $F(\mathrm{id}_1) = \mathrm{id}_B$, and $F(a) = f : A \to B$ for some morphism f in $\mathbf{C}$. Since there are no non-trivial relations or compositions in $\mathbf{2}$ beyond identities, any choice of A, B, f defines a valid functor. Thus, the set $\mathbf{Cat}(\mathbf{2}, C)$ is in bijection with the set of triples $(A, B, f : A \to B)$, which is precisely $Mor(\mathbf{C})$ (as each morphism f determines its domain and codomain). Thus, $\mathbf{Cat}(\mathbf{2}, C) \cong Mor(\mathcal{C})$ via $\alpha_{\mathcal{C}}(F) = F(a)$. Naturality holds as $M(G)(\alpha_{\mathcal{C}}(F)) = G(F(a)) = \alpha_{\mathcal{D}}(G \circ F)$ for a functor $G : C \to D$. Therefore, $M \cong \mathbf{Cat}(\mathbf{2}, -)$ naturally, so M is represented by $\mathbf{2}$.

Exercise 3. Let $\mathbf{C}$ be a category with products and exponentials. You may assume that $\mathbf{C}$ is locally small. Prove that for every object A, we have $A \Rightarrow (B \times C) \cong (A \Rightarrow B) \times (A \Rightarrow C)$. In other notation: $(B \times C)^A \cong B^A \times C^A$. [Hint: one possibility is to use the representable functors definition. You don't need to spell out the proofs that all transformations are natural, but you should convince yourself that they are.]

Since $\mathbf{C}$ has exponentials, it is Cartesian closed, meaning for objects X, Y, there is an exponential object Y^X with evaluation $\text{ev} : Y^X \times X \to Y$. This gives a natural isomorphism $\mathbf{C}(- \times A, B) \cong \mathbf{C}(-, B^A)$. Products give $\mathbf{C}(-, B \times C) \cong \mathbf{C}(-, B) \times \mathbf{C}(-, C)$. To prove the isomorphism, since $\mathbf{C}$ is locally small, by the Yoneda lemma, it suffices to show that $\mathbf{C}(-, (B \times C)^A)$ and $\mathbf{C}(-, B^A \times C^A)$ are naturally isomorphic, implying the objects are isomorphic. Such for any object $D \in \mathbf{C}$,

$$\begin{aligned}
\mathbf{C}(D, (B \times C)^A) &\cong \mathbf{C}(D \times A, B \times C) && \text{(by exponential)} \\
&\cong \mathbf{C}(D \times A, B) \times \mathbf{C}(D \times A, C) && \text{(by product)} \\
&\cong \mathbf{C}(D, B^A) \times \mathbf{C}(D, C^A) && \text{(by exponential)} \\
&\cong \mathbf{C}(D, B^A \times C^A). && \text{(by product)}
\end{aligned}$$

The exponential adjunction is natural by definition. The product isomorphism is natural because precomposition distributes over pairs. Thus, $\mathbf{C}(-, (B \times C)^A) \cong \mathbf{C}(-, B^A \times C^A)$ naturally, so by Yoneda, $(B \times C)^A \cong B^A \times C^A$.

Exercise 4. This exercise will explore exponentials in categories of metric space like things. Given two metric spaces A and B, we can define a metric space $A \multimap B$ where the carrier set is the set of non-expansive functions from A to B, and the metric is the supremum metric:

$$d_{A \multimap B}(f, g) \triangleq \sup_{a \in A} d_B(f(a), g(a)).$$

Note that this supremum can be infinite, so technically we are considering *extended* metric spaces, where the distance function can assign distance ∞.

1. In A1, you showed that the Cartesian product in **Met** is given by the max distance. Explain why it seems difficult to show the isomorphism:

   $$\mathbf{Met}(Z \times X, Y) \cong \mathbf{Met}(Z, X \multimap Y)$$

   In fact, the category **Met** is not Cartesian closed (maybe we will try to show this later) so it is not possible to fix this problem by defining the hom object a bit differently (i.e., with a different metric).

2. Even though an exponential object mentions only two particular objects, its definition involves the entire category. Although it may be counter-intuitive, one way we can make **Met** Cartesian closed is to restrict to a *smaller* category.

Let **UMet** be the subcategory of *ultrametric spaces*, where distances satisfy the stronger condition:

$$d(x,z) \leq \max(d(x,y), d(y,z))$$

Show that we have an isomorphism:

$$\mathbf{UMet}(Z \times X, Y) \cong \mathbf{UMet}(Z, X \multimap Y)$$

3. Another way to make **Met** Cartesian closed is to enlarge the category. Let **Dist** be the category *distance spaces*, which are sets X with a distance function that satisfy (1) $d(x,x) = 0$ and (2) $d(x,y) = d(y,x)$, but may assign distance zero to distinct points and *don't* need to satisfy the triangle inequality. We still consider morphisms in **Dist** to be non-expansive maps.

   Suppose we define a distance space $A \twoheadrightarrow B$ with carrier set being the set of all non-expansive functions from A to B, and distance:

$$d_{A \twoheadrightarrow B}(f,g) \triangleq \sup \{d_B(f(x), g(y)) \mid d_A(x,y) < d_B(f(x), g(y))\}$$

   where the supremum of an empty set is defined to be zero. Show that we have an isomorphism:

$$\mathbf{Dist}(Z \times X, Y) \cong \mathbf{Dist}(Z, X \twoheadrightarrow Y)$$

   where the Cartesian product $X \times Y$ is defined in the same way as in **Met**.

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1. Currying gives a bijection on underlying sets: $f : Z \times X \to Y$ non-exp corresponds to $\bar{f} : Z \to X \multimap Y$, $\bar{f}(z)(x) = f(z,x)$. However, the correspondence must preserve non-expansiveness. If f non-exp, then each $\bar{f}(z)$ is non-exp (fix $z = z'$, bound by d_X). Also, $d(\bar{f}(z), \bar{f}(z')) = \sup_x d_Y(f(z,x), f(z',x)) \leq d_Z(z,z')$ (fix $x = x'$, bound by $\max(d_Z, 0)$). If assume $\bar{f}$ non-exp. Need

$$d_Y(f(z,x), f(z',x')) \leq \max(d_Z(z,z'), d_X(x,x'))$$

   Using triangle inequality in Y:

$$\begin{aligned} d_Y(\bar{f}(z,x), \bar{f}(z',x')) &\leq d_Y(\bar{f}(z,x), \bar{f}(z',x)) + d_Y(\bar{f}(z',x), \bar{f}(z',x')) \\ &\leq \sup_u d_Y(\bar{f}(z,u), \bar{f}(z',u)) + d_X(x,x') \\ &\leq d_Z(z,z') + d_X(x,x') \end{aligned}$$

   But sum can exceed max if both distances are positive and equal. Thus, f may not be non-expansive. Since **Met** is not Cartesian closed, no alternative metric on function spaces yields the adjunction.

2. **UMet** is the full subcategory of ultrametric spaces: $d(x,z) \leq \max(d(x,y), d(y,z))$. Product remains max (preserves ultra). Exponential $X \multimap Y$ is ultra: $d(f,h) = \sup_a d_Y(f(a), h(a)) \leq \max(\sup d(f,g), \sup d(g,h)) = \max(d(f,g), d(g,h))$, by pointwise max from Y. If f non-exp, then each $\bar{f}(z)$ is non-exp (fix $z = z'$, bound by d_X). Also, $d(\bar{f}(z), \bar{f}(z')) = \sup_x d_Y(f(z,x), f(z',x)) \leq d_Z(z,z')$ (fix $x = x'$, bound by $\max(d_Z, 0)$). If assume $\bar{f}$ non-exp,

$$\begin{aligned} d_Y(\bar{f}(z,x), \bar{f}(z',x')) &\leq \max(d_Y(\bar{f}(z,x), \bar{f}(z',x)), d_Y(\bar{f}(z',x), \bar{f}(z',x'))) \\ &\leq \max(\sup_u d_Y(\bar{f}(z,u), \bar{f}(z',u)), d_X(x,x')) \\ &\leq \max(d_Z(z,z'), d_X(x,x')) \end{aligned}$$

   Thus, $\mathbf{UMet}(Z \times X, Y) \cong \mathbf{UMet}(Z, X \multimap Y)$

3. To show the isomorphism $\mathbf{Dist}(Z \times X, Y) \cong \mathbf{Dist}(Z, X \twoheadrightarrow Y)$ naturally in Z, we establish a bijection between non-expansive maps $f : Z \times X \to Y$ and non-expansive maps $\bar{f} : Z \to X \twoheadrightarrow Y$, via currying: $\bar{f}(z)(x) = f(z,x)$. Forward direction: Suppose f is non-expansive. Each $\bar{f}(z) : X \to Y$ is non-expansive as $d_Y(f(z,x), f(z,x')) \leq \max(d_Z(z,z), d_X(x,x')) = d_X(x,x')$. Then, check $\bar{f}$ is non-expansive: $d(\bar{f}(z), \bar{f}(z')) \leq d_Z(z,z')$. The distance is the sup over witnesses where $d_X(u,v) < d_Y(f(z,u), f(z',v))$. For any such witness, $d_Y(f(z,u), f(z',v)) \leq \max(d_Z(z,z'), d_X(u,v))$. Since $d_X(u,v) < d_Y \leq \max$, and $d_X(u,v) < d_X(u,v)$ is impossible if $\max = d_X(u,v)$, it must be $\max = d_Z(z,z')$, so each witness is bounded by $d_Z(z,z')$, hence the sup is too. Reverse direction: Suppose $\bar{f}$ is non-expansive. Need to show f is non-expansive. Suppose that $d_Y(f(z,x), f(z',x')) > \max(d_Z(z,z'), d_X(x,x'))$. If $d_X(x,x') \geq d_Y(...)$, then $d_Y \leq d_X(x,x') \leq \max$, contradiction. If $d_X(x,x') < d_Y(...)$, then $(u = x, v = x')$ is a witness, so $d_Y(...) \leq d(\bar{f}(z), \bar{f}(z')) \leq d_Z(z,z') \leq \max$, again contradiction. Thus, the inequality holds. The bijection is natural in Z because currying commutes with precomposition, preserving the non-expansive condition.

Exercise 5. In this exercise we will explore how to give semantics to if-statements. In order to simplify the presentation we will assume that the type grammar is given by $\tau ::= \mathsf{bool}$ and the raw terms are given by

$$t ::= x \mid \mathsf{true} \mid \mathsf{false} \mid \mathsf{if}\, t\, \mathsf{then}\, t_1\, \mathsf{else}\, t_2$$

with the typing rules

$$\textsc{True}\ \frac{}{\Gamma \vdash \mathsf{true} : \mathsf{bool}} \qquad \textsc{False}\ \frac{}{\Gamma \vdash \mathsf{false} : \mathsf{bool}} \qquad \textsc{If-Then-Else}\ \frac{\Gamma \vdash t : \mathsf{bool} \qquad \Gamma \vdash t_1 : \tau \qquad \Gamma \vdash t_2 : \tau}{\Gamma \vdash \mathsf{if}\, t\, \mathsf{then}\, t_1\, \mathsf{else}\, t_2 : \tau}$$

satisfying the equations:

$$\beta\text{-If-True}\ \frac{\Gamma \vdash t_1 : \tau \qquad \Gamma \vdash t_2 : \tau}{\Gamma \vdash \mathsf{if\ true\ then}\ t_1\ \mathsf{else}\ t_2 = t_1 : \tau}$$

$$\beta\text{-If-False}\ \frac{\Gamma \vdash t_1 : \tau \qquad \Gamma \vdash t_2 : \tau}{\Gamma \vdash \mathsf{if\ false\ then}\ t_1\ \mathsf{else}\ t_2 = t_2 : \tau}$$

Let $\mathbf{C}$ be a category with finite products and a terminal object 1.

1. Assume that bool is interpreted as an object Ω of $\mathbf{C}$ equipped with two distinct morphisms, $T : 1 \to \Omega$ and $F : 1 \to \Omega$, and for every object A, assume that there is a natural transformation $\Phi : \mathbf{C}(-, A) \times \mathbf{C}(-, A) \to \mathbf{C}(- \times \Omega, A)$. Show how T and F can be used to interpret true and false, and how Φ can be used to give semantics to the rule If-Then-Else.

2. Using the semantics of If-Then-Else from the previous part, write down equations which must hold for morphisms so that equations β-If-True and β-If-False hold.

Following what we saw for products, we might expect that the equations imply there is an inverse natural transformation $\Phi^{-1} : \mathbf{C}(- \times \Omega, A) \to \mathbf{C}(-, A) \times \mathbf{C}(-, A)$, which would imply that Ω is not just any old object, but actually the coproduct $1 + 1$, which would make intuitive sense.

But, our equations are not sufficient: they don't assert that the morphism from the if-then-else/case analysis is *unique*. While we needed just a single equation (the η-law) for products to guarantee uniqueness, for coproducts we would need a bunch of somewhat messier equations (*commuting conversions*) to guarantee uniqueness. You can see what these look like on page 80 of Proofs and Types, J.-Y. Girard, Y. Lafont, P. Taylor.

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1. The semantics for True in context $\Gamma \vdash \text{true} : \text{bool}$ is $[[\Gamma \vdash \text{true} : \text{bool}]] = T \circ !_{[[\Gamma]]} : [[\Gamma]] \to \Omega$, where $!_{[[\Gamma]]} : [[\Gamma]] \to 1$ is the unique morphism to the terminal object. Similarly, for False: $[[\Gamma \vdash \text{false} : \text{bool}]] = F \circ !_{[[\Gamma]]} : [[\Gamma]] \to \Omega$. For the If-Then-Else rule in context $\Gamma \vdash \text{if } t \text{ then } t_1 \text{ else } t_2 : \tau$, where $t : \text{bool}$, $t_1, t_2 : \tau$, and $A = [[\tau]]$, the semantics is

   $$[[\Gamma \vdash \text{if } t \text{ then } t_1 \text{ else } t_2 : \tau]] = \Phi_{[[\Gamma]]}([[t_1]], [[t_2]]) \circ \langle \text{id}_{[[\Gamma]]}, [[t]] \rangle$$

   where, $\langle \text{id}_{[[\Gamma]]}, [[t]] \rangle : [[\Gamma]] \to [[\Gamma]] \times \Omega \to A$

   This uses Φ as the eliminator, pairing the branches with the condition.

2. Using the semantics above, the equations that must hold for morphisms so that the β-rules are satisfied are, for any object G and morphisms $f, g : G \to A$,

   $$\Phi_G(f, g) \circ \langle \text{id}_G, T \circ !_G \rangle = f \quad (\beta\text{-If-True}),$$

$$\Phi_G(f, g) \circ \langle \mathrm{id}_G, F \circ !_G \rangle = g \quad (\beta\text{-If-False}).$$

These ensure that when the condition is constantly true or false, the if-expression reduces to the corresponding branch.

Exercise 6 (Category of the Week). Recall that for any category $\mathbf{C}$ and A any object in $\mathbf{C}$, the *slice category* $\mathbf{C}/A$ has as objects $X \xrightarrow{f} A$ in $\mathbf{C}$, and morphisms being commuting triangles. It turns out that slice categories often have nice structure.

1. Prove that for any morphism $f : A \to B$, we have $(C/B)/f \simeq C/A$. That is, the slice of a slice category is still a slice category.
2. Let $\mathbf{C} = \mathbf{Set}$ and let A be any set. In A1, you showed that $\mathbf{Set}/A$ has products. In fact, the slice category is also Cartesian closed. Given objects $X \xrightarrow{f} A$ and $Y \xrightarrow{g} A$, show that the first projection from the set

   $$X \Rightarrow Y \triangleq \{(a, h) \mid a \in A, h : f^{-1}(a) \to g^{-1}(a)\}$$

   is the internal hom in $\mathbf{Set}/A$.

We say that $\mathbf{C}$ is a *locally Cartesian closed category* (LCCC) if every slice category $\mathbf{C}/A$ is Cartesian closed. Locally Cartesian closed categories provide categorical models for dependent type theories, but there are subtle details.

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1. The objects of $(\mathbf{C}/B)/f$ are morphisms in $\mathbf{C}/B$ to the object $f : A \to B$, i.e., $\begin{array}{ccc} Y & \xrightarrow{k} & A \\ & \searrow^{q} & \downarrow f \\ & & B \end{array}$ So $q = f \circ k$, and the object is essentially $k : Y \to A$, matching objects of $\mathbf{C}/A$. Morphisms in $(\mathbf{C}/B)/f$ between two such (k, k') are morphisms in $\mathbf{C}/B$ (i.e., $m : Y \to Y'$ with $q' \circ m = q$, which holds as both $f \circ k'$) commuting over f, i.e., $k' \circ m = k$, exactly morphisms in $\mathbf{C}/A$. The isomorphism functor is the evident one, sending (Y, k) to (Y, k), preserving everything.This shows slice of slice is slice, a key property in LCCCs.
2. Let $\mathbf{C} = \mathbf{Set}$, A any set. $\mathbf{Set}/A$ has objects functions $f : X \to A$ (families of sets $X_a = f^{-1}(a)$), morphisms $h : (X, f) \to (Y, g)$ with $g \circ h = f$ (fiberwise maps). $\mathbf{Set}/A$ has products (fiber products) and is Cartesian closed. Given $f : X \to A$, $g : Y \to A$, the exponential Y^X in $\mathbf{Set}/A$ is the set $\{(a, h) \mid a \in A, h : f^{-1}(a) \to g^{-1}(a)\}$, with projection $\pi : (a, h) \mapsto a : Y^X \to A$. To verify, the internal hom satisfies $\mathbf{Set}/A((Z, r) \times_A (X, f), (Y, g)) \cong \mathbf{Set}/A((Z, r), (Y^X, \pi))$. The product $(Z, r) \times_A (X, f) = \{(z, x) \mid r(z) = f(x)\}$, with map to A by $r(z)$. A morphism to (Y, g)

is a map $k : Z \times_A X \to Y$ with $g \circ k = r \circ \pi_Z$ (over A). Fiberwise, over a, maps $r^{-1}(a) \times f^{-1}(a) \to g^{-1}(a)$. Currying gives maps to Y^X: $\bar{k} : z \mapsto (r(z), h_z)$ where $h_z(x) = k(z, x)$, and since non-expansive is trivial in Set, the bijection holds. The first projection π is indeed the map to A.

Optional Exercises. The initial and terminal object in a category are often denoted by 0 and 1, respectively. It may help to remember that in **Set**, the initial object 0 is the empty set with *zero* elements, and the terminal object 1 is a singleton set with *one* element.

In any category, these objects satisfy the expected laws for products ($A \times 1 \cong A$ and $A \times 0 \cong 0$) and coproducts: ($A + 0 \cong A$). For exponentials, the initial object satisfies $A \Rightarrow 0 \cong 0$ and $0 \Rightarrow A \cong 1$, while the terminal object satisfies $(A \Rightarrow 1) \cong 1$ and $(1 \Rightarrow A) \cong A$. For more practice performing calculations, prove any of these isomorphisms.

Product with Terminal $A \times 1 \cong A$: Assume the category has binary products and a terminal object 1, with unique map $!_C : C \to 1$ for any C. The product $A \times 1$ has projections $\pi_A : A \times 1 \to A$, $\pi_1 : A \times 1 \to 1$. Define $\phi : A \to A \times 1$ as the unique pairing $\langle \mathrm{id}_A, !_A \rangle$, where $!_A : A \to 1$. Define $\psi : A \times 1 \to A$ as π_A. Then, $\psi \circ \phi = \pi_A \circ \langle \mathrm{id}_A, !_A \rangle = \mathrm{id}_A$. For $\phi \circ \psi = \mathrm{id}_{A \times 1}$, note $\mathrm{id}_{A \times 1} = \langle \pi_A, \pi_1 \rangle$. The composite is $\langle \mathrm{id}_A, !_A \rangle \circ \pi_A$. Check projections: $\pi_A \circ (\phi \circ \psi) = \mathrm{id}_A \circ \pi_A = \pi_A$. $\pi_1 \circ (\phi \circ \psi) = !_A \circ \pi_A$. Since maps to 1 are unique, $!_A \circ \pi_A = \pi_1$, as both are the unique map from $A \times 1$ to 1. Thus, projections match, so by uniqueness, $\phi \circ \psi = \mathrm{id}$.

Coproduct with Initial $A + 0 \cong A$: Assume binary coproducts and initial 0, with unique $!_C : 0 \to C$. The coproduct $A + 0$ has inclusions $i_A : A \to A + 0$, $i_0 : 0 \to A + 0$. Define $\phi : A + 0 \to A$ as the unique copair $[\mathrm{id}_A, !_A]$, where $!_A : 0 \to A$. Define $\psi : A \to A + 0$ as i_A. Then, $\phi \circ \psi = [\mathrm{id}_A, !_A] \circ i_A = \mathrm{id}_A$, by UMP. For $\psi \circ \phi = \mathrm{id}_{A+0}$, note $\mathrm{id}_{A+0} = [i_A, i_0]$. The composite is $i_A \circ [\mathrm{id}_A, !_A]$. Check inclusions match: $i_A \circ \mathrm{id}_A = i_A$, $i_A \circ !_A = i_0$ (unique maps from 0).

Exponential with Initial $0^A \cong 1$: Assume exponentials, initial 0, terminal 1. The exponential 0^A satisfies $\mathbf{C}(Z, 0^A) \cong \mathbf{C}(Z \times A, 0)$. Since 0 is initial, $hom(Z \times A, 0) = 1$ if the category has strict initial or by assumption in the problem (as a convention where such isomorphisms hold, per the text). By Yoneda, since $hom(-, 0^A) \cong hom(-, 1)$, then $0^A \cong 1$. Similar fluent proofs apply to the others, like $1^A \cong 1$ (constant functions unique).